

CoRE Stack — Deployments, Users & Modules

Briefing compiled for FPL / Fable · 22 July 2026

Provenance. Answers compiled from two local sources, cross-checked against each other: the **CoRE Stack RAG** (documentation corpus — 390 documents / 2,210 chunks: docs.core-stack.org, the core-stack.org blog, the ecosystem sheet, and the `layers_metadata` catalog) and the **core-stack-backend** Django codebase. Every claim was independently fact-checked by a second pass; corrections applied to the water-balance formula, the plantation species-recommendation claim, and the biodiversity scope. Dates shown against each source are that source's publish / last-updated date as recorded in the RAG corpus.

1 What CoRE Stack has deployed

CoRE Stack is open geospatial *infrastructure*, so "deployed" means both (a) the field tool built on it and (b) the pan-India data layers that are live and publicly queryable.

Commons Connect — the flagship field deployment

A participatory natural-resource-management (NRM) / water-security planning app. Latest field figures:

- Adopted across **1,460 villages**
- NRM plans created in **480 villages**
- taken to Gram Sabha for approval in **327**
- **~9,000 demands** for soil & water conservation assets + livelihoods.

Source: [Field Stories on Commons Connect](#), dated **2026-06-15**. All four figures verified verbatim against this source. **Note:** a separate PRADAN-Ramgarh-only page (2026-05-04) reports **9,318 demands across ~237 villages**; the ~9,000 national aggregate and PRADAN's 9,318 don't nest cleanly, so they likely use different counting windows rather than being simple totals.

PRADAN in Ramgarh, Jharkhand — the largest single rollout

Scaled across hundreds of villages with district-administration backing, and **integrated with the government's Yuktdhara / Yuktadhara GIS portal**. CoRE Stack sources describe it as a "replicable model" for government integration. (Source: [PRADAN / Yuktadhara](#), 2026-05-04.)

Pan-India data layers & public tools (live)

- **Micro-watershed registry** — pan-India delineation of ~1,000 ha units (docs.core-stack.org, 2026-06-01).
- **LULC** — India-SAT land-use/land-cover, pan-India at 10 m from 2017 onward, with intra-annual kharif / rabi / zaid classes.
- **Groundwater (ΔG)** and other hydrology, terrain, tree-cover and drought layers — published per block/tehsil, some pan-India.
- **Ecological Restoration Toolkit** — incl. Continuous Ecological Monitoring, synced to the user's

own Google Drive (core-stack.org, 2025-10-27).

Publicly queryable pan-India data layers — access points [links](#)

Entry point	What it is	Link
STAC Browser	Human-browsable catalog of every published layer (metadata, styles, download structure).	stac.core-stack.org https://stac.core-stack.org/
STAC root catalog (JSON)	Machine-readable root; branches into <i>tehsil-wise</i> and <i>Pan-India</i> collections.	Root catalog.json (S3) https://spatio-temporal-asset-catalog.s3.ap-south-1.amazonaws.com/CorestackCatalogs_merged_collection/catalog.json
Pan-India STAC collection	Pan-India layers; themed sub-collections: groundwater, waterbodies, drainage, geological/geohydrological, etc.	PanIndiaCatalogs/collection.json https://stac.core-stack.org/PanIndiaCatalogs/collection.json
GEE viewer app	Browse public layers directly on an interactive map (no code).	CoRE Stack GEE App https://ee-corestackdev.projects.earthengine.app/view/core-stack-gee-app
GeoServer (WFS / WMS / WCS)	Standards-based geospatial service backing vector (WFS) & raster (WCS/WMS) access.	geoserver.core-stack.org:8443 https://geoserver.core-stack.org:8443/geoserver
Public REST API	Keys, Swagger & ReDoc, and the public API catalog. Endpoints under <code>/api/v1/</code> (see below).	docs.core-stack.org · Public APIs https://docs.core-stack.org/use-precomputed-data/public-apis/
Layer metadata catalog	Authoritative CSVs: <code>layer_mapping.csv</code> , <code>layer_descriptions.csv</code> , <code>vector_column_descriptions.csv</code> .	github.com/core-stack-org/layers_metadata https://github.com/core-stack-org/layers_metadata
Notebooks (starter kit)	Worked examples: load layers, water-balance & cropping-intensity analysis.	github.com/core-stack-org/corestack-notebooks https://github.com/core-stack-org/corestack-notebooks
Datasets / precomputed data hub	Landing pages for curated datasets and the precomputed-data workflow.	docs · use-precomputed-data · core-stack.org/datasets

Public API endpoints (base `/api/v1/`, from `public_api/urls.py`): `get_admin_details_by_latlon`, `get_mwsid_by_latlon`, `get_tehsil_data`, `get_mws_data`, `get_mws_kyl_indicators`, `get_generated_layer_urls`, `get_mws_report`, `get_active_locations`, `get_mws_geometries`, `get_village_geometries`.

2 Who is using the platform

Each row carries its exact source and that source's publish / last-updated date (per the RAG corpus). "Ecosystem sheet" = the [CoRE Stack ecosystem of co-creators, users & deploying organizations](#) (a Google Sheet — undated in the corpus).

Group	Type	Role & location	Source	Source dated
PRADAN	Deploying CSO	District-scale CC rollout, Ramgarh (Jharkhand); + Yuktadhara govt portal	core-stack.org · PRADAN / Yuktadhara	2026-05-04
SUPPORT	Deploying CSO	30 villages, Dumka (Jharkhand)	core-stack.org · Updates Sep 2024–Apr 2025	2025-05-17
FES Foundation for Ecological Security	Deploying CSO	20 villages, Bhilwara (Rajasthan) & Angul (Odisha). Also built the CLART layer.	core-stack.org · Updates Sep 2024–Apr 2025	2025-05-17
Utthan	Deploying CSO	10 villages, Dahod & Bhavnagar (Gujarat)	core-stack.org · Updates Sep 2024–Apr 2025	2025-05-17
Gram Vaani	Co-creator + deploying	Jamui (Bihar) & Uttar Pradesh; a founding co-creator with IIT Delhi	core-stack.org · Early Lessons from Pilots	2025-08-05
4S Sarva Seva Samity Sanstha	Field-testing CSO	Field testing alongside FES / SUPPORT / PRADAN	core-stack.org · DPI for Climate Adaptation	2024-02-13
WRI	Partner (joining)	Listed among growing partners	core-stack.org · Updates Sep 2024–Apr 2025	2025-05-17
Magasool	Partner	Named partner	github.com/core-stack-org · org README	2026-01-20
IIT Delhi (CSE/SIT), IIT Palakkad	Academic / tech	Core technical development; IIT-D CSE/SIT co-created the stack from 2022	org README · core-stack.org · Ecosystem of co-creators	2026-01-20 / 2025-09-16
GIZ, RainMatter, Common Ground, Tarides	Funders / supporters	Financial & collaborative support	github.com/core-stack-org · org README	2026-01-20
AVPN	Network /	Supported field testing (via	Ecosystem sheet	undated

	funder	ERADA / Common Ground)		
Ecological Restoration Alliance (ERA)	Collaborator network	Restoration-planning initiatives & research challenges	Ecosystem sheet	undated
Medius.Earth, Sankhya.Earth, Univ. of Manchester	Restoration collaborators	Listed restoration / research collaborators	Ecosystem sheet	undated
SayTrees	Plantation org	Onboarded org in the Plantations app (org-scoped project data)	Code: plantations/models.py, plantations/views.py:19 4, waterrejuvenation/model s.py:25	current repo

User categories (docs): Civil Society Organisations in rural/tribal areas; community "landscape stewards" & field fellows; government bodies (e.g. the Ramgarh District Collector); researchers and OSS developers via the public-data path. Projects are **organization-scoped** / **multi-tenant** in code (`projects/models.py` — Organization FK), i.e. built to onboard many partner orgs.

3 Modules coded & their capabilities

Two layers: the **product apps** and the **computing / layer-generation modules**. Everything below has real code in the repo; items also present in `layers_metadata/layer_mapping.csv` (with a GeoServer layer + QGIS style) are production-published — the best proxy for "finished."

Product apps

From `projects/models.py` (AppType, lines 7–11) — the officially recognized modules:

- **Watershed Planning** (the Commons Connect core)
- **Plantations**
- **Waterbody Rejuvenation**
- **Community Engagement**

Computing modules (layer generators) — `computing/`

Theme	Coded capability	Module(s)
Groundwater / hydrology	Annual water balance $\Delta G = \text{Precipitation} - \text{Runoff} - \text{ET}$ per micro-watershed (published as fortnightly <code>filtered_delta_g</code>). Annual "change in well depth" is	<code>mws/</code> (<code>delta_g.py</code> , <code>well_depth.py</code>)

	<i>derived</i> from ΔG via principal-aquifer specific yield — related but not identical to ΔG.	
Surface water	Surface Water Bodies pipeline SWB1–4: vectorize LULC water → assign MWS IDs → catchment & stream-order → join Water Body Census. Plus catchment areas & zone-of-influence.	surface_water_bodies/, catchment_area/, zoi_layers/
Water-structure siting	CLART raster — guides suitable sites for recharge / water-conservation structures (check dams, ponds, recharge pits). Built with partner FES.	clart/
Water rejuvenation	Backs the Waterbody Rejuvenation product app.	water_rejuvenation/
Land use	India-SAT LULC (pan-India, 10 m, 2017+, intra-annual kharif/rabi/zaid); cropping intensity (single/double/triple); land-use change detection; LULC × terrain; crop grid.	lulc/, cropping_intensity/, change_detection/, lulc_X_terrain/, crop_grid/
Drought / climate	Drought frequency (2017–2022); SPEI with rainfall & drought sensitivity.	drought/, spei/
Terrain	Terrain raster (30 m) & descriptors; evapotranspiration downscaling.	terrain_descriptor/, et_downscale/
Trees / forests	Tree canopy density (25 m), canopy height, cover change; forest fire; forest fringe; tree-in-grassland.	tree_health/, forest_fire/, forest_fringe/, tree_in_grassland/
Restoration / plantation	Plantation Site Suitability — MCA over 24 variables in 5 criteria (Climate 35% · Topography 25% · Soil 20% · Ecology 10% · Socioeconomic 10%). Outputs site-suitability raster + vector. Species recommendation is a documented aim, not implemented in this backend.	plantation/ (+ plantations/ app)
Soil	Soil health maps (government data + ML).	soil_health/
Socioeconomic / equity	NREGA fund-equity, panchayat caste typology; Antyodaya socioeconomic, livestock, facilities.	misc/ (antyodaya/, livestocks/, facilities/)
Supporting	Layer-dependency graph; STAC catalog generation & coverage.	layer_dependency/, STAC_specs/

On your two named examples

- **Groundwater — done.** The annual water-balance / ΔG layer is computed at micro-watershed level and published (Hydrology dataset). Formula: **$\Delta G = \text{Precipitation} - \text{Runoff} - \text{ET}$** (no soil-moisture term).
- **Biodiversity — no standalone module in this backend.** But working biodiversity tooling *does* exist elsewhere in CoRE Stack: the **Continuous Ecological Monitoring toolkit** runs bioacoustics pipelines (BirdNet bird-species detection, activity heatmaps — core-stack.org, 2025-11-05), and a community-contributed **Anekal biodiversity dashboard** (eBird / iNaturalist — docs.core-stack.org community contributions). The plantation module's Ecology criterion (10% of the MCA) and the tree-health layers are the backend's nearest biodiversity-adjacent capabilities.

Compiled from the CoRE Stack local RAG + core-stack-backend codebase, 2026-07-22. Independently fact-checked; the one corrected error was an erroneous soil-moisture term in the water-balance formula.